Weekly Progress Log – Week 4

Project Title: Data Science with Python
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Week: 4

Objectives for Week

- Understand descriptive statistics and the role of mean, median, mode, range, variance, standard deviation, quartiles, and correlation.
- Study probability basics including probability, conditional probability, normal distribution, and the Central Limit Theorem at an intuitive level.
- Apply descriptive statistics to a real dataset.
- Perform full data cleaning and create a statistical summary using an independent dataset.
- Prepare a Jupyter Notebook showing the cleaning process, descriptive statistics, and a short written summary of findings.

Work Completed

- Prepared Statistics and Probability study notes covering descriptive statistics, measures of central tendency, measures of dispersion, correlation, probability, conditional probability, distributions, normal distribution, sampling concepts, and the Central Limit Theorem.
- Selected the Wine dataset from scikit-learn as the independent dataset for the Week 4 mini project.
- Loaded the Wine dataset and converted it into a pandas DataFrame.
- Inspected the dataset using head(), tail(), shape, columns, and info().
- Confirmed that the dataset contains 178 wine samples and 14 columns, including 13 chemical measurement variables and one target variable.
- Performed data-cleaning checks for missing values, duplicate rows, data types, and minimum/maximum value ranges.
- Confirmed that no missing values or duplicate records were present and created a cleaned DataFrame for analysis.
- Generated a descriptive statistical summary using describe() and reviewed count, mean, standard deviation, minimum, quartiles, median, and maximum.
- Calculated mean, median, mode, variance, and standard deviation for the alcohol variable.
- Created a summary table for alcohol, malic acid, magnesium, color intensity, and proline including mean, median, mode, variance, standard deviation, minimum, and maximum.
- Calculated a correlation matrix for the numerical variables and examined the relationship between alcohol and proline.
- Compared average measurements across the three wine classes and checked class counts.
- Prepared a short-written summary of findings based on the data-cleaning and statistical analysis results.

Challenges Faced

- Understanding the difference between measures of central tendency and measures of dispersion.
- Understanding how variance and standard deviation both measure spread but are expressed differently.

- Understanding how quartiles appear in pandas describe() as 25%, 50%, and 75%.
- Interpreting correlation values correctly and remembering that correlation does not prove causation.
- Deciding which data-cleaning steps were necessary when the selected dataset was already relatively clean.
- Understanding conditional probability, normal distribution, and the Central Limit Theorem at an intuitive level.
- Keeping the mini project focused on the actual deliverable: cleaning, descriptive statistics, and a short-written summary.

Solutions Implemented

- Prepared structured study notes with definitions and examples for the required statistics and probability concepts.
- Separated central-tendency measures (mean, median, mode) from dispersion measures (range, variance, standard deviation) to make interpretation clearer.
- Used pandas describe() to review the overall statistical summary, including quartiles, and calculated mode and variance separately.
- Used explicit missing-value, duplicate, and data-type checks before performing statistical analysis.
- Created clean_df as a cleaned copy of the original DataFrame before continuing the analysis.
- Used a custom statistical-summary table to compare important measures across selected Wine dataset variables.
- Calculated a correlation matrix and a specific alcohol-proline correlation to practice interpretation of relationships between numerical variables.
- Reviewed the assignment requirements and kept the mini project centered on data cleaning, descriptive statistics, and findings rather than adding unnecessary analysis.

Individual Tasks

Responsibility	Description of Contribution	Remark
Statistics Fundamentals	Studied descriptive statistics including mean, median, mode, range, variance, standard deviation, quartiles, and correlation.	Improved understanding of central tendency, dispersion, and relationships between variables.
Probability Fundamentals	Studied probability, conditional probability, normal distribution, sampling concepts, and the Central Limit Theorem at an intuitive level.	Built foundational probability knowledge.
Dataset Selection	Selected the scikit-learn Wine dataset as the independent dataset for the mini project.	New real dataset selected for Week 4 analysis.
Dataset Inspection	Reviewed dataset structure, size, column names, first/last rows, and data types.	Confirmed 178 rows and 14 columns.
Data Cleaning	Checked missing values, duplicates, data types, and minimum/maximum values; created a cleaned DataFrame.	Dataset was already clean; no major corrections were required.
Descriptive Statistics	Generated describe() summary and calculated mean, median, mode, variance, standard deviation, minimum, and maximum for selected variables.	Statistical characteristics of the dataset were summarized.
Correlation Analysis	Calculated the correlation matrix and examined alcohol correlations, including alcohol versus proline.	Practiced interpreting positive, negative, and weak linear

Responsibility	Description of Contribution	Remark
		relationships.
Group Comparison	Calculated average selected measurements by wine class and reviewed class counts.	Compared statistical differences across the three wine classes.
Documentation	Prepared the Jupyter Notebook and short written summary of findings.	Week 4 mini project organized and prepared for submission.